

Millikan's experiment + J.J Thomson exp-8
photon - defn - photo electric effect Threshold frequency
and stopping potential (1)

• Einstein's photo electric eqn (2)

• Graph - 3

• Bohr's postulates (2)

• Energy + radius + velocity deflection - (3)

• law of photoelectric emission - (3)

Mechanics

- Vertical mass spring The ~~simple~~ simple pendulum (3)

- Moment of inertia or radius of gyration (3) - defn

- Angular momentum - conservation principle examples (3)

- md ~~end axis~~ middle axis or rotational (3)

- Rational dynamics - numerical (3)

- Biol - Sacet - Friction - the axis & circular coil (3)

Wave - Newton's formula Laplace correction (3)

- Open and close pipe (3)

- Factors affecting velocity of sound - Temp, pressure, density [Numerical] (2)

Magnetism,

Ampere's Circuital law - defn - Application 3

Solenoid to rod

Force between parallel conductor (3)

$F = IB\ell \sin\theta$ - numerical (2)

HOSTEL EVALUATION TEST 2083

CLASS: 12 'CHEMISTRY'

TIME: 2 HOURS

FULL MARK: 51

GROUP 'A'

Short Questions [5X5=25+2]

1. Write down chemistry of Blue vitriol. [5]
2. What is rusting? Describe the electrochemical theory of rusting of iron. List any two methods for the prevention of rusting. [5]
3. An aliphatic haloalkane (A) heated with aqKOH to give (B) reacts with SOCl_2 to produce (C) The compound further reacts with AgCN to yield (D) with LiAlH_4 produces (E) The compound (a) undergoes wurtz reaction to give an alkane of molecular formula C_4H_{10} . Identify (A), (B), (C), (D) and (E). Also write the reactions involved. [5]
4. a) Chlorobenzene is less reactive than chloroethane towards nucleophilic substitution reaction. why? [2]
b) What happens when. [3]
 - i) Chlorobenzene is heated with Ni-Al in alkaline medium
 - ii) Chlorobenzene is heated with chloral in acidic medium
 - iii) Chlorobenzene is heated with chloromethane in presence of Na-metal
5. Differentiate between primary standard and secondary standard solution. 3g of a trivalent metal was completely dissolved with 750ml of 1N HCl. The residual solution further required 100ml of N/2 NaOH for complete neutralization. Find the atomic mass of metal. [2+3]
[2]
6. Define the following term. [2]
 - i) End point
 - ii) Acidimetry

GROUP 'B'

Long Questions: [8X3=24]

7. An organic compound (A) reacts with acetone to produce a kind of sleep-inducing drug.
 - i) How would you prepare compound (A) by using one isomer of $\text{C}_2\text{H}_6\text{O}$. [2]
 - ii) Why is compound (A) stored in an air-tight bottle by adding small amount of ethanol? [2]
 - iii) What happens when compound (A) is: [4]
 - a) heated with conc. HNO_3 ?
 - b) reduced in acidic medium?
 - c) boiled with aq. alkali?
 - d) heated with phenol in presence of aq. NaOH?
8. a) Write the applications of solubility product principle and common ion effect in qualitative salt analysis. [3]
b) Derive relationship between p^{H} and p^{OH} . [2]
c) State and explain ostwald's dilution law. Also mention its limitation. [3]
[3]
9. a) Give example of each of the following Rxn:
 - i) Williamson's ether synthesis;
 - ii) Sandmeyer's Rxn.
 - iii) Fittig Rxn.
b) Distinguish ethanol, propan-2-ol and 2-methylpropan-2-ol by victormeyer's method. [4]
c) A chemist kept pentan-2-one and pentan-3-one in two different beaker but forgot to label. How will you help him to distinguish it. [1]

Best of Luck

HOSTEL EVALUATION TEST 2083

Grade: XI

Subject: Physics

Time: 2 hours

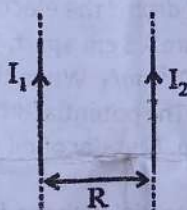
Full Marks: 51

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

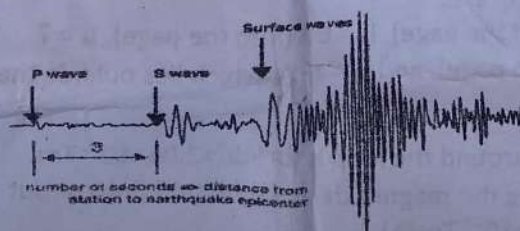
Attempt all the questions.

Group A

1. (a) If a simple pendulum is taken to the Moon, will it gain or lose time? Why? [1]
(b) Find an expression for the energy of a particle in simple harmonic motion (SHM) and show that the particle obeys the law of conservation of energy. [2]
(c) A body of mass 2 kg is suspended from a spring of negligible mass and is found to stretch the spring by 0.1 m. Find its force constant and time period. [2]
2. (a) State the principle of conservation of angular momentum. [1]
(b) If the Earth were to shrink suddenly, what would happen to the length of the day? Give a reason. [2]
(c) A constant torque of 500 Nm turns a wheel that has a moment of inertia of 20 kgm² about its centre. Find the angular velocity gained in 2 seconds and the kinetic energy gained. [2]
3. Two conductors X and Y, carrying currents I_1 and I_2 , are placed in air at a distance r apart, as shown in the figure below.



- (i) Find the magnitude of the force between them. [2]
(ii) What is the nature of the force in this case? [1]
(iii) What is the nature of the force when the currents flow in opposite directions? [1]
(iv) Define one ampere of current on the basis of the force between parallel current-carrying conductors. [1]
4. (a) What is the photoelectric effect? [1]
(b) Photons come from three lamps that emit red, green, and blue light. Which of these lamps produces photons with the highest energy? [2]
(c) A clean nickel surface with a work function of 5.1 eV is exposed to light of wavelength 235 nm. What is the maximum speed of the photoelectrons emitted from its surface? [2]
5. (a) Why was Newton's assumption about the velocity of sound in a gaseous medium wrong? [1]
(b) Discuss the correction made by Laplace for the velocity of sound in a gaseous medium. [2]



- (c) An earthquake occurs at a certain distance from a seismic station. The P-wave travels with a velocity of 8 km/s and the S-wave travels with a velocity of 4.5 km/s. If the S-wave arrives 10 s after the P-wave, find the distance of the earthquake from the seismic station. [2]
6. (a) Derive the progressive wave equation. [2]

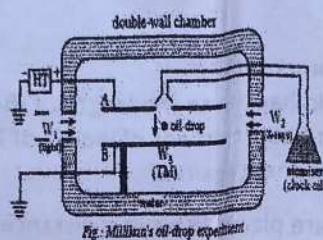
(b) A progressive wave is given by $y = 0.03 \sin(3\pi t - 0.03\pi x)$, where x is in metres and t is in seconds. Calculate the phase difference between two particles 0.05 metre apart at any instant. Also, find the maximum amplitude of the resultant wave when this wave superposes with another wave. [2+1]

7. (a) Derive an expression for the rotational kinetic energy of a rigid body rotating with angular velocity ω , and hence define moment of inertia. [2+1]

(b) A ballet dancer spins at 2.4 rev/s with her arms outstretched, when the moment of inertia about the axis of rotation is I . With her arms folded, the moment of inertia about the same axis becomes $0.6I$. Calculate the new rate of spin. [2]

Group B

8. (a) The figure below shows the experimental setup of Millikan's oil drop experiment. Find the expression for the charge of an oil drop of radius r moving with constant velocity v in the downward direction, using a free body diagram. [2]



(b) What will be the expression for the charge of an oil drop if the electric force is greater than its weight? [1]

(c) In a Millikan-type apparatus, the horizontal plates are 1.5 cm apart. With the electric field switched off, an oil drop is observed to fall with a steady velocity of 2.5×10^{-2} cm/s. When the electric field is switched on, with the upper plate positive, the drop just remains stationary. The potential difference between the plates is 1500 V. Calculate the number of electronic charges on the drop. (Density of oil = 900 kg/m^3 and viscosity of air = $1.8 \times 10^{-5} \text{ Ns/m}^2$; neglect air density.) [2]

(d) Sketch graphs showing the variation of stopping potential with the frequency of incident radiation for three photosensitive materials A, B, and C, having threshold frequencies f_1 , f_2 , and f_3 respectively, such that $f_1 > f_2 > f_3$. [1]

(i) In which case is the stopping potential greater, and why? [1]

(ii) Does the slope of the graph depend on the nature of the material used? Explain. [1]

9. (a) An electron moves northward in a magnetic field directed vertically upward. What is the direction of the magnetic force on it? [1]

(b) State Ampere's circuital law. [1]

(c) Using Ampere's circuital law, derive an expression for the magnetic field at a point on the axis of an infinitely long current-carrying solenoid. [3]

(d) In the figure below, several long straight conductors carrying currents pass perpendicular to the plane of the page. A closed Amperian loop is drawn as shown.

The currents inside and outside the loop are:

$I_1 = 12 \text{ A}$ (into the page), $I_2 = 3 \text{ A}$ (out of the page), $I_3 = 6 \text{ A}$ (into the page), $I_4 = ?$ (direction unknown), $I_5 = 35 \text{ A}$ (into the page), and $I_6 = 15 \text{ A}$, which lies outside the loop.

The line integral of the magnetic field around the loop is $\oint \vec{B} \cdot d\vec{l} = 2.05 \times 10^{-5} \text{ Tm}$.

Using Ampere's circuital law, determine the magnitude and direction (into or out of the page) of the current I_4 . ($\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$)

